

Chapter 2: Review of Probability and Statistics

R Exercises (with Solutions)

1. Problem

Simulating a discrete random variable

The number of customer complaints W that a small company receives per day has the probability mass function

w_k	0	1	2	3	4
$p_W(w_k)$	0.10	0.25	0.30	0.20	0.15

- Compute $E(W) = \sum_k w_k p_W(w_k)$ and $\text{Var}(W) = \sum_k (w_k - E(W))^2 p_W(w_k)$ in R using the vectors `w <- 0:4` and `p <- c(0.10, 0.25, 0.30, 0.20, 0.15)`.
- Use `sample()` to simulate the number of complaints on $n = 1000$ days.
- Estimate the pmf by relative frequencies and compare with the true pmf. Draw a bar plot of the relative frequencies with `barplot()`.
- Compute the sample mean and the sample variance of your simulated data and compare them with the true values from (a).
- Repeat (b)–(d) with $n = 20$ days. What do you observe?

Solution

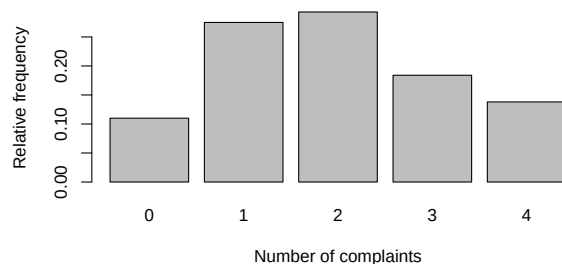
```
(a) > w <- 0:4
> p <- c(0.10, 0.25, 0.30, 0.20, 0.15)
> EW <- sum(w * p)
> VarW <- sum((w - EW)^2 * p)
> c(EW, VarW)
[1] 2.0500 1.4475

(b) > n <- 1000
> w_sim <- sample(w, size = n, replace = TRUE, prob = p)

(c) > p_hat <- table(factor(w_sim, levels = w)) / n
> rbind(p_hat, p)

      0      1      2      3      4
p_hat 0.11 0.275 0.293 0.184 0.138
p      0.10 0.250 0.300 0.200 0.150

> barplot(p_hat, xlab = "Number of complaints", ylab = "Relative frequency")
```



The relative frequencies are close to the true probabilities.

(d) `> mean(w_sim)`

```
[1] 1.965
```

`> var(w_sim)`

```
[1] 1.451226
```

Both are close to $E(W) = 2.05$ and $\text{Var}(W) = 1.4475$.

(e) `> w_sim20 <- sample(w, size = 20, replace = TRUE, prob = p)`

`> table(factor(w_sim20, levels = w)) / 20`

```
  0    1    2    3    4
0.15 0.25 0.30 0.15 0.15
```

`> mean(w_sim20)`

```
[1] 1.9
```

`> var(w_sim20)`

```
[1] 1.673684
```

With only $n = 20$ days, the estimates are much further away from the true values, and they change a lot from simulation to simulation. The larger n , the more precise the estimates (Law of Large Numbers).

2. Problem

Densities in R

- The delay X (in hours) of a parcel delivery has the density $f_X(x) = x/2$ for $0 \leq x \leq 2$ and $f_X(x) = 0$ otherwise. Use the R function `integrate()` to check that f_X integrates to one, and to compute $P(0.5 \leq X \leq 1.5)$ and $E(X) = \int x f_X(x) dx$.
- Plot the density f_X with `curve()` for $x \in [-0.5, 2.5]$.
- Generate $n = 200$ realizations of a normal random variable with mean 1.6 and variance 2 using `rnorm()`.
- Estimate the density of your realizations with a histogram (`hist(..., freq = FALSE)`) and with a kernel density estimator (`density()`). Add the true density using `curve()` and `dnorm()`.
- Repeat (c) and (d) with $n = 20$ and with $n = 5000$. What do you observe?

Solution

(a) `> f_X <- function(x) ifelse(x >= 0 & x <= 2, x / 2, 0)`

`> integrate(f_X, lower = 0, upper = 2)$value` # total probability

```
[1] 1
```

`> integrate(f_X, lower = 0.5, upper = 1.5)$value` # $P(0.5 \leq X \leq 1.5)$

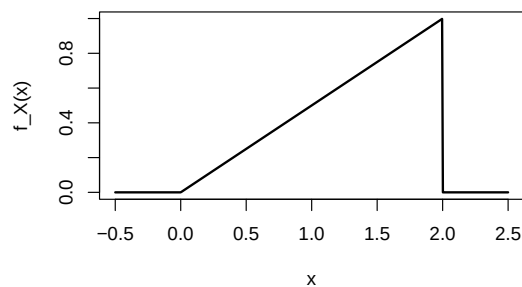
```
[1] 0.5
```

`> integrate(function(x) x * f_X(x), lower = 0, upper = 2)$value` # $E(X)$

```
[1] 1.333333
```

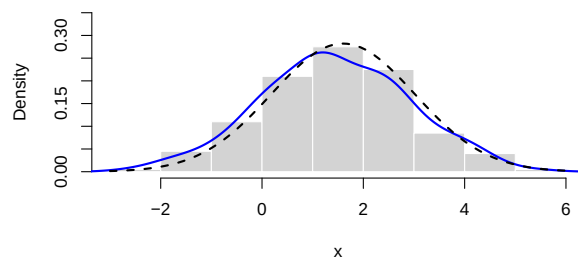
(b) `> curve(f_X(x), from = -0.5, to = 2.5, n = 501,`

`+ xlab = "x", ylab = "f_X(x)", lwd = 2)`



```
(c) > n <- 200
    > x <- rnorm(n, mean = 1.6, sd = sqrt(2)) # sd = square root of the variance

(d) > hist(x, freq = FALSE, main = "", col = "lightgray", border = "white",
    +     ylim = c(0, 0.35))
    > lines(density(x), lwd = 2, col = "blue") # kernel density estimate
    > curve(dnorm(x, mean = 1.6, sd = sqrt(2)), add = TRUE, lwd = 2, lty = 2) # true density
```



(e) With $n = 20$, the histogram and the kernel density estimate are rough and can be quite far from the true density. With $n = 5000$, both estimates are very close to the true density: the estimates converge to the true density as $n \rightarrow \infty$.

3. Problem

Empirical cdf, sample moments, covariance and correlation

A retail chain records the advertising expenditures X (in 1000 EUR) and the sales Y (in 1000 units) of $n = 10$ of its stores:

$X_{i,\text{obs}}$	7	3	5	4	6	8	2	5	9	1
$Y_{i,\text{obs}}$	25	18	21	22	24	27	17	23	26	16

- Store the data in a data frame with the two columns advertising and sales.
- Compute the sample means, the sample variances and the sample standard deviations of both variables with `mean()`, `var()` and `sd()`.
- Compute the sample variance of X “by hand” in R, i.e., using only `sum()`, `length()` and `mean()`, and check that you get the same result as with `var()`.
- Compute and plot the empirical cdf of the sales with `ecdf()`. Use it to estimate the probability that a store sells more than 22,000 units.
- Draw a scatter plot of the data, and compute the sample covariance and the sample correlation with `cov()` and `cor()`.

- (f) Convert advertising to EUR and sales to units, and recompute the covariance and the correlation. Explain the result.

Solution

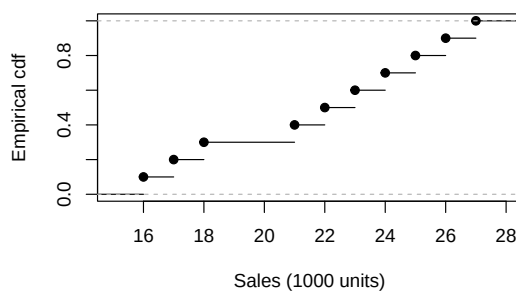
```
(a) > stores <- data.frame(advertising = c(7, 3, 5, 4, 6, 8, 2, 5, 9, 1),
+                           sales       = c(25, 18, 21, 22, 24, 27, 17, 23, 26, 16))

(b) > mean(stores$advertising); var(stores$advertising); sd(stores$advertising)
[1] 5
[1] 6.666667
[1] 2.581989

> mean(stores$sales); var(stores$sales); sd(stores$sales)
[1] 21.9
[1] 14.76667
[1] 3.842742

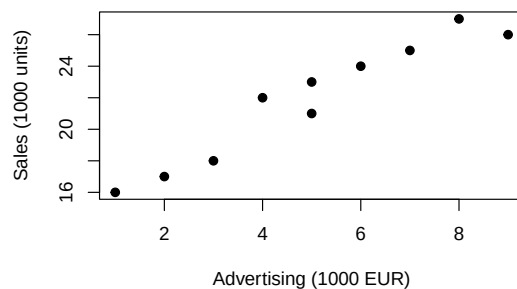
(c) > x <- stores$advertising
> sum((x - mean(x))^2) / (length(x) - 1)
[1] 6.666667

(d) > F_hat <- ecdf(stores$sales)
> plot(F_hat, main = "", xlab = "Sales (1000 units)", ylab = "Empirical cdf")
> 1 - F_hat(22) # estimate of P(Y > 22)
[1] 0.5
```



```
(e) > plot(stores$advertising, stores$sales, pch = 19,
+         xlab = "Advertising (1000 EUR)", ylab = "Sales (1000 units)")
> cov(stores$advertising, stores$sales)
[1] 9.555556

> cor(stores$advertising, stores$sales)
[1] 0.9630755
```



```
(f) > cov(1000 * stores$advertising, 1000 * stores$sales)
```

```
[1] 9555556
```

```
> cor(1000 * stores$advertising, 1000 * stores$sales)
```

```
[1] 0.9630755
```

The covariance is multiplied by $1000 \cdot 1000 = 10^6$, since $\text{Cov}(a_X X, a_Y Y) = a_X a_Y \text{Cov}(X, Y)$; it depends on the units of measurement. The correlation does not change, because it standardizes the covariance by the standard deviations.

4. Problem

The normal distribution in R

A machine fills coffee into cups. The filled amount X (in ml) is normally distributed with $\mu = 200$ and $\sigma = 5$.

- Use `pnorm()` to compute $P(X < 190)$ and $P(195 \leq X \leq 205)$.
- Use `qnorm()` to find the amount x that is undercut by only 5% of all cups.
- Standardize: compute $P(X < 190)$ a second time using `pnorm()` with its default arguments `mean = 0` and `sd = 1`.
- Simulate 10,000 fillings with `rnorm()`. Estimate $P(X < 190)$ and the 5% quantile from the simulated values (*hint*: `mean()` of a logical vector and `quantile()`), and compare with (a) and (b).
- Plot the densities of $\mathcal{N}(200, 5^2)$, $\mathcal{N}(210, 5^2)$ and $\mathcal{N}(200, 10^2)$ in one plot using `curve()` and `dnorm()`. How do μ and σ change the shape?

Solution

```
(a) > pnorm(190, mean = 200, sd = 5)
```

```
[1] 0.02275013
```

```
> pnorm(205, mean = 200, sd = 5) - pnorm(195, mean = 200, sd = 5)
```

```
[1] 0.6826895
```

```
(b) > qnorm(0.05, mean = 200, sd = 5)
```

```
[1] 191.7757
```

```
(c) > pnorm((190 - 200) / 5)
```

```
[1] 0.02275013
```

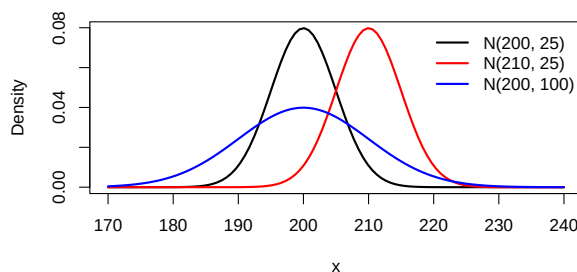
```
(d) > x_sim <- rnorm(10000, mean = 200, sd = 5)
```

```
> mean(x_sim < 190) # relative frequency of cups below 190 ml
```

```
[1] 0.025
> quantile(x_sim, 0.05)      # empirical 5% quantile
5%
191.6701
```

Both are close to the exact values from (a) and (b).

```
(e) > curve(dnorm(x, mean = 200, sd = 5), from = 170, to = 240,
+         lwd = 2, xlab = "x", ylab = "Density")
> curve(dnorm(x, mean = 210, sd = 5), add = TRUE, lwd = 2, col = "red")
> curve(dnorm(x, mean = 200, sd = 10), add = TRUE, lwd = 2, col = "blue")
> legend("topright", legend = c("N(200, 25)", "N(210, 25)", "N(200, 100)"),
+       col = c("black", "red", "blue"), lwd = 2, bty = "n")
```



Changing μ shifts the bell curve (location); a larger σ makes it wider and flatter (spread).

5. Problem

Chi-squared and F-distribution by simulation

Let $Z_1, \dots, Z_4$ be independent standard normal random variables and

$$W = Z_1^2 + Z_2^2 + \dots + Z_4^2 \sim \chi_4^2.$$

- Why can W never take negative values? What are $E(W)$ and $\text{Var}(W)$?
- Use R to generate 10,000 realizations of W *only* with the help of `rnorm()`.
Hint: `matrix(rnorm(10000 * 4), ncol = 4)` generates a matrix with 10,000 rows and 4 columns; `rowSums()` sums up each row.
- Compute the sample mean and the sample variance of your realizations and compare them with the values from (a).
- Draw a histogram of your realizations (`freq = FALSE`) and add the density of the χ_4^2 distribution using `curve()` and `dchisq()`.
- Estimate $P(W > 9.49)$ from your simulated realizations, and compare with the exact value from `pchisq()`.
- Now generate 10,000 realizations of

$$V = \frac{W_1/4}{W_2/20}, \quad W_1 \sim \chi_4^2, \quad W_2 \sim \chi_{20}^2, \quad W_1 \perp W_2,$$

using `rchisq()`. Draw a histogram and add the density of the $F_{4,20}$ distribution (`df()`).

Solution

- W is a sum of squares, and squares are never negative. $E(W) = \nu = 4$ and $\text{Var}(W) = 2\nu = 8$.

```
(b) > nu <- 4
> Z <- matrix(rnorm(10000 * nu), ncol = nu) # 10000 x nu standard normals
> W <- rowSums(Z^2) # sum of squares in each row
```

```
(c) > mean(W) # compare with E(W) = nu
```

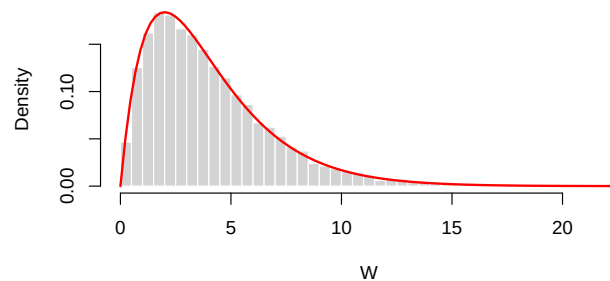
```
[1] 4.040859
```

```
> var(W) # compare with Var(W) = 2 * nu
```

```
[1] 7.867581
```

Both values are close to $\nu = 4$ and $2\nu = 8$, respectively.

```
(d) > hist(W, breaks = 50, freq = FALSE, main = "",
+       col = "lightgray", border = "white")
> curve(dchisq(x, df = nu), add = TRUE, lwd = 2, col = "red")
```



```
(e) > mean(W > 9.49) # estimate from the simulation
```

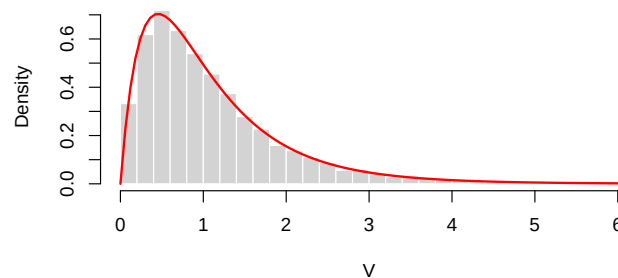
```
[1] 0.0503
```

```
> 1 - pchisq(9.49, df = nu) # exact value
```

```
[1] 0.04995313
```

Both values are close to 0.05 (9.49 is the 95% quantile of the χ_4^2 distribution).

```
(f) > nu2 <- 20
> W1 <- rchisq(10000, df = nu)
> W2 <- rchisq(10000, df = nu2)
> V <- (W1 / nu) / (W2 / nu2)
> hist(V, breaks = 80, freq = FALSE, main = "", xlim = c(0, 6),
+       col = "lightgray", border = "white")
> curve(df(x, df1 = nu, df2 = nu2), add = TRUE, lwd = 2, col = "red")
```



In both cases, the histogram of the simulated realizations matches the theoretical density.

6. Problem

Law of Large Numbers

Let D denote the result of rolling a fair die, $D \in \{1, 2, 3, 4, 5, 6\}$, with $E(D) = 3.5$.

- (a) Use R to simulate $n = 1000$ rolls of a fair die with `sample()`.
- (b) Compute the running sample means

$$\bar{D}_{m,\text{obs}} = \frac{1}{m} \sum_{i=1}^m D_{i,\text{obs}} \quad \text{for } m = 1, 2, \dots, 1000,$$

and plot them against m . Add a horizontal line at $E(D) = 3.5$.

Hint: `cumsum(d) / seq_along(d)` computes all running means at once.

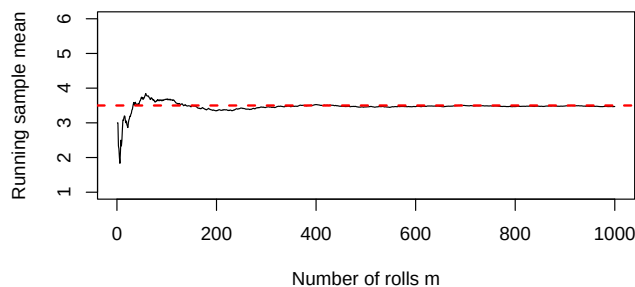
- (c) Repeat (a) and (b) a few times. What changes from repetition to repetition, and what does not?
- (d) Now consider a loaded die with $P(D = 6) = 0.5$ and $P(D = k) = 0.1$ for $k = 1, \dots, 5$. Compute $E(D)$ for this die by hand. Repeat (a) and (b) for the loaded die and check your result.

Solution

- (a)

```
> n <- 1000
> d <- sample(1:6, size = n, replace = TRUE)
> head(d)
[1] 3 3 1 2 1 1
```
- (b)

```
> running_mean <- cumsum(d) / seq_along(d)
> plot(running_mean, type = "l", ylim = c(1, 6),
+      xlab = "Number of rolls m", ylab = "Running sample mean")
> abline(h = 3.5, lty = 2, col = "red", lwd = 2)
```

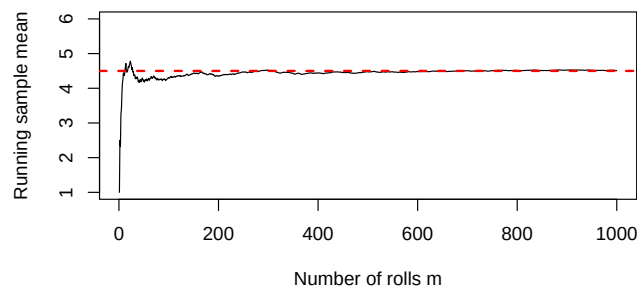


- (c) The path of the running means changes in every repetition; in particular, for small m the running means fluctuate strongly. But in every repetition, the running mean settles down close to $E(D) = 3.5$ as m gets large. This is the Law of Large Numbers: $\bar{D}_{m,\text{obs}} \rightarrow E(D)$ as $m \rightarrow \infty$.

- (d)

$$E(D) = (1 + 2 + 3 + 4 + 5) \cdot 0.1 + 6 \cdot 0.5 = 1.5 + 3 = 4.5.$$

- ```
> d_loaded <- sample(1:6, size = n, replace = TRUE,
+ prob = c(0.1, 0.1, 0.1, 0.1, 0.1, 0.5))
> plot(cumsum(d_loaded) / seq_along(d_loaded), type = "l", ylim = c(1, 6),
+ xlab = "Number of rolls m", ylab = "Running sample mean")
> abline(h = 4.5, lty = 2, col = "red", lwd = 2)
```



The running means now settle down close to 4.5.

## 7. Problem

### Simulating from a data-generating process

The monthly electricity costs  $Y_i$  (in EUR) and the living space  $X_i$  (in  $\text{m}^2$ ) of households are generated by the DGP

$$Y_i = \beta_1 + \beta_2 X_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2),$$

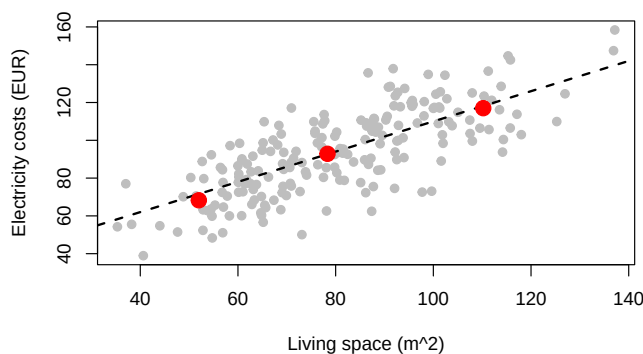
with  $\beta_1 = 30$ ,  $\beta_2 = 0.8$ ,  $\sigma = 15$ , and  $X_i \sim \mathcal{N}(80, 20^2)$  independent of  $\varepsilon_i$ .

- Simulate  $n = 200$  households from this DGP.
- Draw a scatter plot and add the true regression line  $\beta_1 + \beta_2 x$  with `abline()`.
- Split the simulated households into three groups:  $X_i \leq 60$ ,  $60 < X_i \leq 100$  and  $X_i > 100$ . Compute the mean of  $X_i$  and the mean of  $Y_i$  within each group and add these three points to your plot. What do you observe?
- Compute the sample correlation between  $X_i$  and  $Y_i$ . Then repeat the simulation with  $\sigma = 40$ . How do the scatter plot and the correlation change?

### Solution

- ```
> beta1 <- 30; beta2 <- 0.8; sigma <- 15
> n <- 200
> X <- rnorm(n, mean = 80, sd = 20)
> eps <- rnorm(n, mean = 0, sd = sigma)
> Y <- beta1 + beta2 * X + eps
```
- The plot below also contains the group means from (c).


```
> plot(X, Y, pch = 19, col = "gray",
+       xlab = "Living space (m^2)", ylab = "Electricity costs (EUR)")
> abline(a = beta1, b = beta2, lwd = 2, lty = 2)
> ## group means (alternatively: mean(Y[X <= 60]), etc.)
> group <- cut(X, breaks = c(-Inf, 60, 100, Inf))
> mean_X <- tapply(X, group, mean)
> mean_Y <- tapply(Y, group, mean)
> points(mean_X, mean_Y, pch = 19, col = "red", cex = 1.8)
```



(c) The group means are computed in the code of (b):

```
> cbind(mean_X, mean_Y)
      mean_X mean_Y
(-Inf,60]  51.99160 68.26145
(60,100]   78.33291 92.85615
(100, Inf] 110.21466 117.00226
```

The three group means (red points) lie close to the true regression line: the group means of Y_i estimate the conditional means $E(Y_i | X_i = x)$ at the group means x of X_i .

(d) `> cor(X, Y)`

```
[1] 0.7508498
```

```
> Y_noisy <- beta1 + beta2 * X + rnorm(n, mean = 0, sd = 40)
```

```
> cor(X, Y_noisy)
```

```
[1] 0.3793239
```

With $\sigma = 40$, the points scatter much more around the (same) regression line, and the correlation is clearly smaller. The regression function $E(Y_i | X_i = x)$ is unchanged; only the noise is larger.

8. Problem

Simulating a confounder

Consider again the DGP for ice cream sales X_i and swimming pool visitors Y_i , which are both driven by the daily maximum temperature Z_i :

$$\begin{aligned} Z_i &\sim \mathcal{N}(22, 5^2), \\ X_i &= 2 + 0.5 Z_i + u_i, & u_i &\sim \mathcal{N}(0, 1), \\ Y_i &= 1 + 0.3 Z_i + v_i, & v_i &\sim \mathcal{N}(0, 1), \end{aligned}$$

with Z_i , u_i and v_i independent of each other.

- Simulate $n = 3650$ days (ten years) from this DGP.
- Draw a scatter plot of (X_i, Y_i) and compute the sample correlation.
- Now only keep the days with a maximum temperature between 24 and 26 degrees Celsius. How many days are left? Compute the sample correlation between X_i and Y_i for these days and compare with (b).
- Color the points in your scatter plot from (b) by temperature group ($Z_i < 18$, $18 \leq Z_i \leq 26$, $Z_i > 26$). What do you see?

Solution

```
(a) > n <- 3650
> Z <- rnorm(n, mean = 22, sd = 5)
> X <- 2 + 0.5 * Z + rnorm(n)
> Y <- 1 + 0.3 * Z + rnorm(n)
```

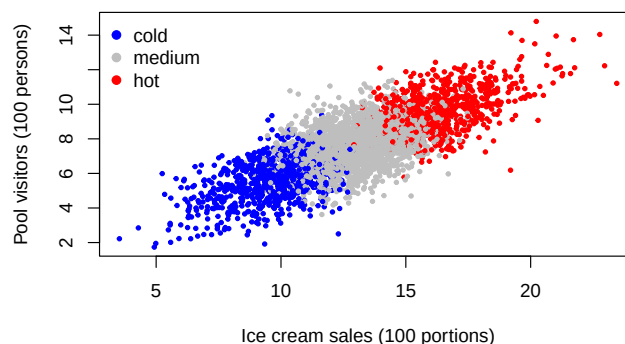
```
(b) > cor(X, Y)
[1] 0.7733173
```

The correlation is strongly positive, although Y_i does not depend on X_i in the DGP. The plot is shown in (d).

```
(c) > similar_temp <- Z >= 24 & Z <= 26
> sum(similar_temp)
[1] 479
> cor(X[similar_temp], Y[similar_temp])
[1] 0.1404151
```

For days with (almost) the same temperature, the correlation is close to zero: once we hold the confounder Z_i (approximately) fixed, there is (almost) no association left between X_i and Y_i .

```
(d) > temp_group <- cut(Z, breaks = c(-Inf, 18, 26, Inf))
> plot(X, Y, pch = 19, cex = 0.5,
+      col = c("blue", "gray", "red")[temp_group],
+      xlab = "Ice cream sales (100 portions)",
+      ylab = "Pool visitors (100 persons)")
> legend("topleft", legend = c("cold", "medium", "hot"),
+      col = c("blue", "gray", "red"), pch = 19, bty = "n")
```



Cold days (blue) have low ice cream sales *and* few pool visitors; hot days (red) have high values of both. Within each temperature group, the point cloud shows much less of an upward trend. The overall positive association is created by the temperature.